



Languages and Peoples of the Eastern Himalayan Region (LPEHR) comprises an annual special issue of *Himalayan Linguistics* together with occasional special publications. Peer-reviewed and open-access, it focuses on linguistic and ethnographic documentation and description in the Eastern Himalaya.

Languages and Peoples of the Eastern Himalayan Region (LPEHR)

Some preliminary notes on Challow, a Trans-Himalayan language from Manipur, India

Jessica K. Ivani

University of Zurich

ABSTRACT

This contribution introduces Challow, a previously unreported and undescribed Trans-Himalayan language spoken in the Ukhrul district of Manipur, North-East India. It includes preliminary sociolinguistic, phonological, and lexical information, compared with data from other languages of the region. The study also offers a 100-word Swadesh list of Challow lexical entries. At first glance, Challow reveals interesting properties, such as the presence of an original *p(ə)*- prefix so far unattested in the languages of the area and typologically rare phonological processes.

KEYWORDS

Tibeto-Burman, Manipur, phonology, lexicon

This is a contribution from *Himalayan Linguistics* Vol 22(2) – Languages and Peoples of the Eastern Himalayan Region: 1-17.

ISSN 1544-7502

© 2024. All rights reserved.

This Portable Document Format (PDF) file may not be altered in any way.

Tables of contents, abstracts, and submission guidelines are available at escholarship.org/uc/himalayanlinguistics

*Some preliminary notes on Challow, a Trans-Himalayan language from Manipur, India*¹

Jessica K. Ivani

University of Zurich

1 Introduction

Challow is a Trans-Himalayan language spoken in the Ukhrul district of Manipur state, India. It is spoken in the villages of Challow and Poi in northern Ukhrul by approximately 2000² speakers who belong to the Tangkhul Naga Scheduled Tribe. Speakers report no mutual intelligibility with the languages spoken in the neighboring villages of Kongai (Kongai language, Ivani, this issue), Huishu (Huishu language, Mortensen 2004), and New Tusom (Suansu language, Ivani 2019a, 2019b, 2022, 2023a).

Challow is unreported and unattested in the linguistic literature. The data currently available on the languages of the Ukhrul district include basic comparative materials, such as wordlists and brief sketches, and more detailed diatopic descriptions of Tangkhul, the lingua franca of the district (Arokianathan 1987, Devi 2011, Leisan 2017, among others). Early attestations on the languages of the region comprise lexical data on Champhung (Brown 1837), Pettigrew's description (1918) of Tangkhul Naga, and the comparative work on 'Naga'³ languages by Marrison (1967). More recently, besides what already mentioned above, we have linguistic outlines of the region (Haokip 2011) and, most notably, the substantial contribution of Mortensen and colleagues on Ukhrul, Kachai, East Tusom, and Sorbung (Mortensen 2003, 2004; Mortensen & Keogh 2011; Mortensen & Miller 2013; Mortensen & Picone 2021; Mortensen et al. 2021). Shared phonological and lexical features across these languages allowed Mortensen to propose the existence of a Tangkhulic language subgroup (Mortensen 2003, Post & Burling 2017), which 'fits' several languages of the district (but not all of them, notably Suansu, Ivani 2023a, Mortensen & Picone 2021; and Sorbung, Mortensen 2011).

Challow is an endangered language, due to lack of documentation and written tradition, the language dominance of Tangkhul Naga, and population dispersal caused by migration and endogamic marriage practices. The data used in this study was collected from a native Challow speaker in his early twenties, living in Maharashtra.

The goal of this contribution is to introduce Challow by providing initial insights on its

¹This study was funded by the Department of Comparative Language Science (IVS) at the University of Zurich, Switzerland, and the Swiss Society for Endangered Languages (Grant type 1). I gratefully acknowledge their support here. I am grateful to the Challow, Kongai, and Suansu speaking communities for their help and encouragement. I thank Grégoire Hansen, Dagmar Jung, Kellen Parker van Dam, Taras Zakharko, as well as the reviewers, for their valuable feedback during the various stages of this work. I am solely responsible for any imprecision.

²The population breakdown is as follows: Challow 351 inhabitants, Poi 1595. Source: 2011 Indian Census (<https://www.census2011.co.in>).

³I use here Naga as a broad ethnic term with no linguistic connotation.

syllable (Section 2.1) and related properties (such as initials and rhymes, Sections 2.2 - 2.3), ‘frozen’ prefixes (Section 2.4), and lexicon (Section 3) from a comparative perspective. The study concludes with a short summary (Section 4). The Challow 100-word Swadesh list is reported in the Appendix, and the related high-quality audio files are available online.⁴ The languages of the Ukhrul district used for the lexical comparison include Champhung, Suansu, Huishu, East Tusom, Kachai and Tangkhul Naga. With the exception of Suansu, the linguistic data from these languages has been extracted from STEDT. The results of this study have been shared and discussed with members of the Tangkhul Naga community before finalizing the write-up of the paper, ensuring their involvement in the research process.

2 Phonological properties

2.1. Syllable structure

A preliminary representation of the Challow syllable is presented in Figure 1. This schema partially reflects and adopts the symbolic conventions of the Proto Trans-Himalayan (hence PTH) syllable canon presented in Matisoff (2003:10; there referred to as Proto Tibeto-Burman).

$$\begin{array}{c} [T] \\ (C_2) C_1 V (C_f) \end{array}$$

Figure 1 – Preliminary representation of Challow syllable

Challow, similarly to many other Trans-Himalayan languages of the area and beyond (Matisoff 2003) is characterized by the presence of sesquisyllabic syllables, or ‘one syllable and a half’ long forms (Matisoff 2003). These ‘half syllables’ refer to ‘frozen’ prefixes, often stacked one after the other and attached to the first syllable of the lexical root. The nucleus of these prefixes is often a centralized vowel that undergoes reduction, hence the ‘half’ status. These forms are briefly discussed in Section 2.4.

The Challow data available does not provide instances of glides or liquids following the consonant initial (C_1) root, found in the PTH syllable canon (Matisoff 2003:10). The vocalic nucleus (V) is followed by a restricted set (C_f) of consonants or semivowels (usually treated as consonants in this position, see Matisoff 2003). Consonant clusters are rare in the data collected so far. This study will not deal with tone; however, an impressionistic account of the data supported by comparative materials suggests that Challow has contrastive lexical tone (T) at the syllable level.⁵

Challow lexical roots include monosyllabic, disyllabic, and polysyllabic units. Challow syllables consist of the vowel nucleus, with either a consonant onset (C_1) or a glide following the nucleus. The (so far) sparse instances of consonant clusters appear to be the result of vowel reduction processes in the stacked frozen prefixes. This is shown, for example, by forms such as *krai.pə* ‘big’, which can be analyzed as *kǎ.rai.pə*, where *kǎ* is the prefix found on all reported forms of verbs and property concepts in Challow (as well as Tangkhul and other neighboring languages). A similar pattern can be observed in *pro.puk* ‘woman’, where the frozen prefix undergoing vowel reduction is most likely *pǎ-*, found extensively on Challow

⁴The Zenodo repository is the following: Ivani, Jessica K. 2023b. Challow 100-word Swadesh list. Zenodo. <https://doi.org/10.5281/zenodo.8338134>.

⁵To account for this uncertainty, I have enclosed the tone symbol in brackets ([T]) in Figure 1.

nouns (see 2.4 below). Other consonant clusters do not seem to be related to prefix stacking processes, for example the plosive plus fricative cluster in *pǎ.pfu.le* (cf. PTH *pwu ‘belly’), which supports the presence of CCV clusters in Challow.

Overall, CV syllables appear to be the most frequent in the sample at hand (accounting for more than 70%, or 152 of the 212 individuated syllables), followed by CVC structures (52 syllables out of 212, including glides and semivowels), VC (such as *aj* ‘first person pronoun’) and the CCV cluster reported above. What follows covers initials, rhymes, and ‘frozen’ prefixes.

2.2. Initials

The initial consonants in the small sample of Challow lexical roots include obstruents, nasals, and liquids.⁶ A preliminary comparison of Challow initials with synchronic data from neighboring languages and reconstructed Proto Tangkhulic (hence PTh) forms shows the relative stability of these forms. In what follows, obstruents are grouped into plosives (Table 1), fricatives (Table 2), and affricates (Table 3). Nasals are illustrated in Table 4, and liquids are shown in Table 5.⁷ Exemplificatory lexemes occur after each Table.

	PTh	Challow	Huishu	Tusom	Suansu	Kachai	Tangkhul	Ukhrul	Champhung
(1)	*k	k		k	k	k	k	k	k
(2)	*p ^h	p ^h	p ^h	p ^h	p ^h	p ^h	p ^h	p ^h	p ^h
(3)	*t ^h	t ^h	t ^h	t ^h	θ	t ^h	t ^h	t ^h	

Table 1 – obstruent plosives initials

- (1) ‘head’, PTh *kow; Challow *pǎ-kao*, Tusom *ʔǎ-kfu*, Suansu *kə*, Kachai *ʔa-ke*, Tangkhul *ā-kui*, Ukhrul *ʔa-kui*, Champhung *kau*
- (2) ‘mountain’, PTh *p^huŋ; Challow, Huishu, Tusom, Suansu, Kachai, Tangkhul, Ukhrul, Champhung *kə-p^huŋ*
- (3) ‘three’ PTh *t^hum; Challow, Huishu *kǎ-t^ham*, Suansu *kə-θam*, Kachai and Ukhrul *kə-t^hum*, Tusom *ka-t^hō*, Tangkhul *thum*

	PTh	Challow	Huishu	Tusom	Suansu	Kachai	Tangkhul	Ukhrul	Champhung
(4)	*w	v	w	w	x	w	v	v	
(5)	*s	s	s	s	s	s	s	s	s
(6)	*ʃ	ʃ	s	s	s	s	ʃ	ʃ	z
(7)	*h	h	h		ɸ	h	f	f	v

Table 2 – obstruent fricative initials

- (4) ‘bird’ PTh *wa; Challow *p^hǎ-va*, Huishu *ʔa-p^h-we*, Kachai *wu-ðu*, Tangkhul and Ukhrul *va-naw*, Champhung *ŋu-t^he* (cf. PTh *ta ‘bird’), Suansu *xe*

⁶This set is not exhaustive. I narrow the scope of this Section to those lexical roots (in the Challow 100-word Swadesh list) that show clear cognates across the languages of the Ukhrul district with reconstructed Proto Tangkhulic forms.

⁷The Tangkhulic protoforms as well as the data from the languages of the Ukhrul district are extracted from the STEDT database (<https://stedt.berkeley.edu>), except for Suansu data (collected from Ivani 2019b, Ivani 2023a). Empty fields indicate missing datapoints. The forms are reported verbatim.

- (5) ‘long’ PTK *saŋ; Challow and Huishu *kǎ-si*, Suansu *a-sǎe*, Kachai *kǎ-so*, Tangkhul and Ukhul *ka-sang*, Tusom *kǎ-suiǎ*
- (6) ‘blood’ PTK *fi; Challow *pǎ-fi*, Huishu *ʔa-sik*, Suansu *a-su*, Kachai *ʔa-sv*, Tangkhul and Ukhul *ʔa-fi*, Tusom *ǎn-su*, Champhung *a-zi*
- (7) ‘dog’ PTK *hwi; Challow *tʰo-huk*, Suansu *mi*, Huishu *ʔa-huk*, Kachai *ʔa-hwi*, Tangkhul *hu*, Tusom *y*, Ukhul *fu*, Champhung *a-val*

	PTk	Challow	Huishu	Tusom	Suansu	Kachai	Tangkhul	Ukhul
(8)	*ts	ts	ts	ts	t	ð	tʃ	ts
(9)	*tʃ	tʃ	ts	tɕ	dʒ		ʃ	ʃ

Table 3 – obstruent affricates initials

- (8) ‘black’ PTK *tsik; Challow *kǎ-tsa-pǎ*, Huishu *kǎ-tsoʔ*, Suansu *a-tǎ-ke*, Kachai *kʰǎ-tseʔk*, Tusom *kǎ-zu*, Ukhul *kǎ-tsik*, Tangkhul *ka-chik*
- (9) ‘hear’ PTK *tʃa; Challow *kǎ-tʃa*, Huishu *tse-kʰǎ-low*, Kachai *kʰǎ-ŋǎ-nv* (cf. PTK *na), Suansu *a-dʒue*, Tangkhul *sha*, Tusom *kǎ-tɕa*, Ukhul *kǎ-fa*

	PTk	Challow	Huishu	Tusom	Suansu	Kachai	Tangkhul	Ukhul	Champhung
(10)	*m	m	m	m	m	m	m	m	m
(11)	*n	n	n	n	n	n	n	n	n

Table 4 – nasal initials

- (10) ‘eye’ PTK *mit; Challow *pǎ-mak*, Huishu *ʔa-moʔ*, Kachai *ʔa-mvʔk*, Tangkhul *mik*, Ukhul *ʔa-mik*, Champhung *a-mak*, Tusom *ʔu-mu*, Suansu *mok*
- (11) ‘ear’ PTK *na; Challow *pǎ-kǎ-ni*, Huishu *ʔa-kʰǎ-ni*, Suansu *ni-kǎ*, Kachai *kʰǎ-nv*, Tangkhul and Ukhul *khǎ-na*, Tusom *ʔi-kʰǎ-ni*, Champhung *kʰu-nu*

	PTk	Challow	Huishu	Tusom	Suansu	Kachai	Tangkhul	Ukhul	Champhung
(12)	*l	l	l		l	l	l		l
(13)	*r	r	r	r	r	r	r	r	r

Table 5 – liquid initials

- (12) ‘earth’ PTK *lej; Challow *nǎ-laj*, Huishu *ʔa-nǎ-lu*, Suansu *la*, Kachai *ŋǎ-li*, Tangkhul *ŋǎ-rəy*, Champhung *ŋa-lai*
- (13) ‘bone’ PTK *ru; Challow *pǎ-ruk*, Huishu *ʔa-ruk*, Kachai *ʔa-rv*, Tangkhul *ā-ra*, Tusom *ʔu-ru-kuǎ*, Ukhul *ʔa-ru-kuj*, Champhung *so-ru*, Suansu *a-ru*

2.3. Rhymes

Challow rhymes include consonants and semivowels, which are usually treated as consonants in the PTH syllable canon (Matisoff 2003). For brevity, I illustrate and discuss only the consonant codas. Table 6 summarizes the consonant rhyme patterns found so far in the data.

V	C _f
a	-m
a	-n
u	-ŋ
a	
u	-k
ɔ	
o	
a	-t

Table 6 – consonant codas

The Challow data exhibits nasals and plosives in the coda position. Nasals [m] and [n] occur after [a], e.g., *kǎt^ham* ‘three’ and *pǎman* ‘name’, while [ŋ] occurs after [u], e.g., *t^hanruŋ* ‘tree’. Regarding plosives, [t] appears to be the most restricted: it is found only after [a], *sekǎt^hat* ‘kill’. Velar plosive [k] is found with [u] *pǎɯuk* ‘bone’, [a] *pǎmak* ‘eye’, [ɔ] *pamǎkɔk* ‘knee’, and [o] *pǎrɔktao* ‘neck’.

Dorsal stop codas are also found in Huishu, where it has been claimed that these epenthetic consonants have emerged later, as the result of a typologically rare and aberrant sound change process (Mortensen 2004).⁸ This process is apparently motivated by aerodynamic, acoustic, and perceptual factors, whereas formal and semantic facts do not play a role (Mortensen 2004: 292). In Huishu, the dorsal stops have been added after high vowels in open syllables with a striking regularity (Mortensen 2004: 294). The same process can be observed in Challow, where dorsal stops are additionally found after low vowels (cf. *kǎnak* ‘two’, *kǎt^hak* ‘die’) and the process seems less regular (cf. *pǎfi* ‘blood’, *pǎnunu* ‘breast’), suggesting a possibly ongoing phonological development. Examples are illustrated in Table 7.

⁸The same sound change has been attested in other languages of NE India, i.e., Chakpa (Huziwara 2020).

	PTH	PTk	Challow	Huishu	Tangkhul	Kachai	Suansu
‘blood’	*shywəy	*a.ʃi	pəʃi	ʔa-sik	āshi	ʔa-sə	asuu
‘die’	*səy	*kətʰi	kətʰak	kə-tik	kəʰthi		atswe
‘one’		*kəsi	kəsǎka	kə-sik-a	khə	kə-sə	səkʰa
‘two’	*g-nis	*kəni	kǎnak	kʰə-nik	khə³ni	kʰə-nə	kəniga
‘salt’	*tsyɪ	*məci	matsak	ʔa-mə-tsik	mətsi	mə-tsə	matsə
‘bone’	*g-rus	*ʔru	pəruk	a-ruk	a°ra	ʔa-rə	aɹə
‘breast’	*nəw	*ʔnu	pənunu	ʔa-nənuk	nu	nə-tə	nwi
‘dog’	*kwəy	*hwi	tʰohuk	ʔa-huk	hu	ʔa-hwi	ɹwi
‘water’	*rəy	*ri	ʃəruk	ʔa-ruk	ru	tuŋ-ði	ʃwi

Table 7 – final dorsal stops in Challow (in bold face) and in the languages of the Ukhrul district

2.4. Prefixes

As mentioned in Section 2.1, Challow lexical roots are characterized by the presence of several ‘frozen’ prefixes with a reduced vowel nucleus and no identifiable semantic content,⁹ whose status (presence, loss, reprefixation processes, and distribution) is of fundamental importance in Trans-Himalayan comparative studies and reconstruction (Matisoff 2003: 87 and references therein). Challow’s linguistic neighbors, and most notably Tangkhul Naga, are known for extensive prefix stacking processes and the use of a “favorite prefix” (Matisoff 2003: 94 and Mortensen 2004), *kʰ(ʰ)(ə)-*, added to all the verb roots. Although these traits are not found in all the languages of the region (cf. notably Suansu, Ivani 2023a), these are attested and widespread enough to include *k(h)(ə)-* among the defining properties of the Tangkhulic language subgroup (Mortensen 2003).

Challow verbs (including adjectives, which do not form a separate class in Challow) consistently add the prefix *kʰ(ʰ)(ə)-*, to the respective roots. Table 8 illustrates some examples, along with the corresponding forms from other languages from the Ukhrul district and the reconstructed PTK forms.

	PTk	Challow	Huishu	Tangkhul	Kachai	Suansu
‘black’	*tsik	kəpʰuŋ	kə-tsoʔ	ka°chik	kʰə-ts ək	atəke
‘cold’	*kow	kəsak	kə-mə-kuŋ	ka°sik	kʰə-mə-ke	mokunje
‘drink’	*maŋ	kəmi		cay	kʰə-ma	amœ
‘eat’	*tsa	kədzə	kə-tse	kəʰtsa		aðe

Table 8 – distribution of *kʰ(ʰ)(ə)-* prefix in Challow (in bold face) and in the languages of the Ukhrul district

Besides *kʰ(ʰ)(ə)-*, Challow prefixes include, pending further data, the formatives *m(ə)-* and *p(ə)-*, although it is worth noting that each of these forms might not trace back to a single form, and might correspond to distinct formatives that got merged in their diachronic paths. Further

⁹Given these properties, in the Trans-Himalayan (Tibeto-Burman) linguistic tradition these forms are often discussed within the phonology, as part of the syllable canon (cf. Matisoff 2003).

studies will hopefully disentangle the histories of these formatives. Given the limited scope of this introduction, only the basic formal and semantic properties of these prefixes are discussed.

The prefix *m(ə)-* is found on verbs: *kʰə.mə.kaj*, ‘to bite’, *kǎ.mǎ.ŋə.wa* ‘to be white’, as well as some body parts, such as *pǎ.mə.luŋ* ‘heart’, *pa.mǎ.kək* ‘knee’, *pǎ.mǎ.tʰan* ‘liver’, and *pǎ.mǎ.laj* ‘tongue’. This formative is most likely older, given its position closer to the root, and the correspondence with the respective PTH roots: **m-lun* ‘heart’, **m-kay* ‘to bite’. This prefix is found in several languages of the Ukhrul district, as illustrated in Table 9.

	PTk	Challow	Huishu	Tangkhul	Kachai	Suansu
‘bite’	*kej	kʰə.mə.kaj	kə-mə-ri	khə¹mə³kəi	kʰə-mə-ki	mkaj
‘heart’	*luŋ	pǎ.mə.luŋ		məluŋ	mə-luŋ	məluŋ
‘liver’	*tʰin	pǎ.mǎ.tʰan	ʔa-mə-tʰen	aᵐmaᵐthin	ʔa-mə-tʰen	atʰē
‘white’	*car	kǎ.mǎ.ŋə.wa		kaᵐchar	kə-tear	aŋoi

Table 9 – distribution of *m(ə)-* prefix in Challow (in bold face) and in the languages of the Ukhrul district

The most striking Challow feature so far is perhaps the pervasive presence of the formative *p(ə)-*, added to most Challow nouns available in the data. This prefix occurs on all lexemes pertaining body parts, including non-human body parts such as *pǎ.kǎ.maj* ‘tail’, *pǎ.nǎ.tʰu* ‘horn’, and *pə.ve* ‘feather’, as well as on lexemes such as *pǎ.man*, ‘name’. This prefix is not found in the other surveyed languages of the Ukhrul district. First evidence suggests that *p(ə)-* is most likely a later development given its position to the lexical root when stacked with older prefixes such as *m(ə)-*, as seen above (Table 9) with body parts lexemes. Table 10 illustrates some examples.

	PTk	Challow	Huishu	Tangkhul	Kachai	Suansu
‘feather’		pə.ve	ʔa-pʰwi-vej	vanoha		aɾe
‘horn’	*ci	pǎ.nǎ.tʰu	ʔa-nə-tsik	angaᵐchi		atsə
‘name’	*miŋ	pǎ.man	ʔa-məŋ	¹ə¹miŋ	ʔa-məŋ	amuŋ
‘tail’	*mej	pǎ.kǎ.maj	ʔa-kʰə-mu	¹ə¹khə²məi		ama

Table 10 – distribution of *p(ə)-* prefix in Challow (in bold face) and in the languages of the Ukhrul district

3 Lexicon

The Challow 100-word Swadesh list is reported in the Appendix.¹⁰ The Tangkhulic language subgroup, to which several languages of the Ukhrul district belong (with some notable exceptions, such as Suansu and Sorbung), is based on phonological and lexical innovations found in the region. Lexical innovations deemed as Tangkhulic include a handful of lexical items. These appear to be adopted by Challow as well, as illustrated in Table 11.¹¹

¹⁰Empty fields indicate unavailable data points. The related audio files are available on Zenodo (<https://doi.org/10.5281/zenodo.8338134>).

¹¹The lexical innovations deemed as Tangkhulic also include the word for mushroom, PTK *waar. This datapoint is not currently available for Challow.

	PTk	Challow	Huishu	Tangkhul	Kachai	Suansu
‘mountain’	*phuŋ	kəp ^h uŋ	ʔa.kə.phuŋ	ka.phuŋ	kə.phuŋ	kə.phuŋ
‘fish’	*khaj	keisə	ʔa.khejfe	khai	khaj	kiθe
‘give’	*kə.mi	k ^h əmi	khə.me	kə.mi	khə.mə	ʔa.mie
‘hand’	*pan	pəpi	ʔa.vej	pang	ʔa.pon	ke
‘foot’	*pej	pəpae	ʔa.phu	phai	ʔa.phi	pedzə

Table 11 – Tangkhulic lexical innovations (Mortensen 2004) and corresponding entries in Challow and the languages of the Ukhrul district

This short study introduced Challow, an unreported language from North-East India until now. Challow shares several Tangkhulic features, such as the prefix *k^(h)(ə)*- added to the verb roots and several lexical innovations. At the same time, Challow shows interesting features, notably the presence of the prefix *p(ə)*- added to many nouns, especially body parts, and typologically rare sound changes, such as the addition of epenthetic consonants to open syllables.

References

- Arokianathan, S. 1987. *Tangkhul Naga Grammar*. Vol. 16 (CIIL Grammar series). Mysore: Central Institute of Indian Languages.
- Brown, Nathan. 1837. Comparison of Indo-Chinese languages. *Journal of the Asiatic Society of Bengal* 6.1023-1038.
- Devi, Thokchom Lakhipriya. 2011. *The Tangkhul grammar of Shangshak village*. Can- chipur: Manipur University dissertation.
- Haokip, Pauthang. 2011. The languages of Manipur: a case study of the Kuki-Chin languages. *Linguistics of the Tibeto- Burman Area* 34(1).
- Huziwara Keisuke. 2020. On the genetic position of Chakpa within Luish languages. *Himalayan Linguistics*, 19(2).
- Ivani, Jessica K. 2019a. A first overview on Suansu, a Tibeto-Burman language from Northeastern India. Paper presented at *the 29th Meeting of the Southeast Asian Linguistics Society (SEALS 29)*, Tokyo University of Foreign Studies, Japan 27–29 May 2019. 34.
- Ivani, Jessica K. 2019b. Suansu 250 concepts wordlist. Zenodo. DOI: 10.5281/zenodo.3383006
- Ivani, Jessica K. 2022. Comparative constructions in Suansu and the languages of northeastern India. *Linguistic Typology at the Crossroads* 2(1). 65–93.
- Ivani, Jessica K. 2023a. Suansu language from northeastern India: A field report. *Linguistics of the Tibeto-Burman Area* 46(1).
- Ivani, Jessica K. 2023b. Challow 100-word Swadesh list. Zenodo. <https://doi.org/10.5281/zenodo.8338134>.
- Leisan, Apinao Reisangmi. 2017. *Grammar and Texts of Hunphun variety of Tangkhul Naga language*. University of Poona dissertation.
- Marrison, Geoffrey E. 1967. *The classification of the Naga languages of north-east India*. School of Oriental & African Studies dissertation.
- Matisoff, James A. 2003. *Handbook of Proto-Tibeto-Burman: System and Philosophy of Sino-*

- Tibetan reconstruction*. Vol. 135. Berkeley & Los Angeles: University of California Press. xlii+750.
- Mortensen, David. 2003. *Comparative Tangkhul*. University of California, Berkeley, Unpublished Manuscript.
- Mortensen, David. 2004. The emergence of dorsal stops after high vowels in Huishu. In *Annual meeting of the Berkeley linguistics society*, vol. 30, 292–303.
- Mortensen, David & Jennifer Keogh. 2011. Sorbung, an undocumented language of Manipur: its phonology and place in Tibeto-Burman. *JSEALS* 4. 64–114.
- Mortensen, David & Jordan Picone. 2021. East Tusom. a phonetic and phonological sketch of a largely undocumented Tangkhulic language. *Linguistics of the Tibeto- Burman Area* 44(2). 168–196.
- Mortensen, David R. & James A. Miller. 2013. A reconstruction of Proto-Tangkhulic rhymes. *Linguistics of the Tibeto- Burman Area* 36(1). 1–32.
- Mortensen, David R., Jordan Picone, Xinjian Li & Kathleen Siminyu. 2021. Tusom2021: A phonetically transcribed speech dataset from an endangered language for universal phone recognition experiments. *CoRR* abs/2104.00824.
- Pettigrew, William. 1918. *Tangkhul [Naga] grammar and dictionary*. Shillong: Assam Secretariat Printing Office. 490.
- Post, Mark W. & Robbins Burling. 2017. The Tibeto- Burman languages of northeast India. *The Sino-Tibetan Languages*. 213–242.

Appendix: Challow 100-word Swadesh list

ALL	k ^h ampan
ASHES	
BARK	
BELLY	papfule
BIG	kraipə
BIRD	pəva
BITE	k ^h əməkaj
BLACK	kətsapə
BLOOD	pəʃi
BONE	pəɽuk
BREAST	pənunu
BURNING	kətʃao
CLOUD	maoʃe
COLD	kəsak
COME	ɽeɽo
DIE	kət ^h ak
DOG	t ^h ohuk
DRINK	kəmi
DRY	kətsu
EAR	pəkəni
EARTH (SOIL)	nəɽaj
EAT	kədzə

EGG	pəpruk
EYE	pəmak
FEATHER	pəve
FIRE	ʃamaj
FISH	keisə
FLY (V)	kəpɛ
FOOT	pəpae
FULL	kəpao
GIVE	kʰəmi
GOOD	kəje
GREASE	
GREEN	kəmkepa
HAIR	pəkaove
HAND	pəpi
HEAD	pəkao
HEAR	kətʃa
HEART	pəməluŋ
HORN (ANATOMY)	pənətʃu
HOT	kətʃao
I	ai
KILL	səkətʰat
KNEE	paməkək
KNOW (SOMETHING)	kətʰaj
LEAF	tʰani
LIEDOWN	pikasə
LIVER	pəmətʰan
LONG	kəsi
LOUSE	ʃərak
MALE PERSON	mpəlak
MANY	ajaka
MEAT	pəsa
MOON	təle
MOUNTAIN	kəpʰuŋ
MOUTH	pəmansu
CLAW OR NAIL	pəpimatan
NAME	pəman
NECK	pəʔoktao
NEW	kətʰa
NIGHT	
NOSE	pənesu
ONE	kəsəka
PERSON	
RAIN (PRECIPITATION)	kənaru
RED	kəmapə
ROAD	keraj
ROOT	pənətʰə
ROUND	
SAND	nəlajsakti
SAY	
SEE	kətʰae

SEED	t ^h ǎt ^h ajpǎmǎtʃi
SIT	
SKIN	pǎfə
SLEEP	kǎpi
SMALL	kǎtiva
SMOKE (EXHAUST)	maikǎfə
STAND	kǎnǎdzan
STAR	mǎpetʃe
STONE	ʃǎluŋ
SUN	tsǎmak
SWIM	
TAIL	pǎkǎmaj
THAT	tʃete
THIS	hai
THOU	no
THREE	kǎt ^h am
TONGUE	pǎmǎlaj
TOOTH	pǎʃǎve
TREE	t ^h anruŋ
TWO	kǎnak
WATER	ʃǎruk
WE (INCLUSIVE)	apai
WHAT	napə
WHITE	kǎmǎŋəwa
WHO	t ^h unde
WOMAN	propuk
YELLOW	